Quiz: 15

(C12) $x = \sqrt{y} - 2y \qquad 1 \le y \le 4$

$$L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

$$L = \int_1^4 \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

$x = \sqrt{y} - 2y$

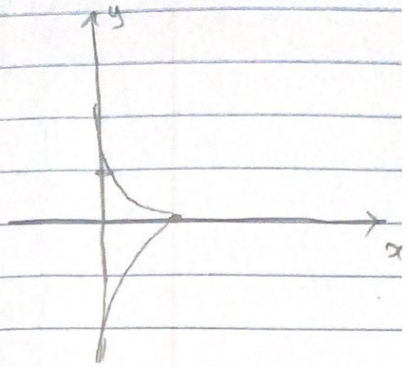
$$\left[\frac{dx}{dy}\right] = \left[\frac{1}{2\sqrt{y}} - 2\right] \Rightarrow \left(\frac{dx}{dy}\right)^2 = \left[\frac{1}{2\sqrt{y}} - 2\right]^2$$

$$\left(\frac{dx}{dy}\right)^2 = \left[\frac{1}{4y} - \frac{2}{\sqrt{y}} + 4\right]$$

$$L = \int_1^4 \sqrt{1 + 4 + \frac{1}{4y} - \frac{2}{\sqrt{y}}}\, dy$$

$$\boxed{L = \int_1^4 \left(\sqrt{5 + \frac{1}{4y} - \frac{2}{\sqrt{y}}}\right) dy}$$

(13) $y = e^{-8x}$, $x \geq 0$ rotated about x-axis




$$SA = \int_a^b 2\pi \,(\text{radius})\,(\text{length})\,dx$$

$$SA = \int_0^{\infty} 2\pi \left(e^{-8x}\right) \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$y = e^{-8x}, \quad \frac{dy}{dx} = e^{-8x}(-8)$$

$$\left(\frac{dy}{dx}\right)^2 = 64\left(e^{-8x}\right)^2 = 64\, e^{-64x^2}$$

$$SA = \int_0^{\infty} 2\pi \left(e^{-8x}\right) \sqrt{1 + 64\, e^{-64x^2}}\, dx$$

$$SA = \int_0^{\infty} 2\pi \,\frac{1}{e^{8x}} \sqrt{1 + \frac{64}{e^{64x^2}}}\, dx$$

$$SA = \int_0^{\infty} (2\pi)\,\frac{1}{e^{8x}} \sqrt{\frac{e^{64x^2} + 64}{e^{64x^2}}}\, dx$$

$$SA = \int_0^{\infty} (2\pi) \left(\frac{1}{e^{8x}}\right) \left(\frac{1}{e^{8x}}\right) \sqrt{e^{64x^2} + 64}\, dx$$

$$SA = \int_0^{\infty} (2\pi)\left(e^{-8x}\right)\sqrt{1 + 64e^{-64x^2}}\,dx$$

Let $u = e^{-8x}$, $du = e^{-8x}(-8)\,dx$

$$\frac{du}{-8} = e^{-8x}\,dx$$

$$SA = \int_0^{\infty} 2\pi\left(\frac{du}{-8}\right)\sqrt{1+u}$$

$$SA = \int_0^{\infty} \left(\frac{2\pi}{-8}\right)\sqrt{1+u}\,du$$

Let $a = 1+u$, $du = da$

$$SA = \int_0^{\infty} \left(\frac{2\pi}{-8}\right)\sqrt{a}\,da$$

$$SA = \frac{2\pi}{-8}\left[\frac{a^{3/2}}{3/2}\right]_0^{\infty}$$

$$SA = \left(\frac{2\pi}{-8}\right)\left(\frac{2}{3}\right)\left[\left(\sqrt{1+e^{-8x}}\right)^3\right]_0^{\infty}$$

$$SA = \left(-\frac{\pi}{12}\right)\left[\right.$$